Learning-Rate Schedules Are Not Adiabatic: A Rate-Dependent Correction for Loss-Curve Prediction

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Peking University Deep Learning Course Task 2
github.com/wjjpku/DL-final

Self-contained summary

MPL and FSL are strong *adiabatic* predictors: they model the quasi-static loss implied by the LR history. This project identifies the missing transient: when LR is annealed faster than the optimizer's excess-loss state can relax, the true loss remains **above** the quasi-static prediction. I derive a low-capacity correction, test it on public LLM curves, and reproduce the mechanism in a real $\sim$10M transformer and controlled AdamW simulations.

Standalone report: project goal, method, experimental settings, results, analysis, limitations, and next steps.

Roadmap

Task 2 coverage checklist

This deck is organized to explicitly cover the six required components.

Requirement	Where it appears in this deck
Problem background and goal	Project Goal section: schedule-aware loss-curve prediction, motivation, and the core question.
Data processing and experimental setup	Data and Reproduction: data source, preprocessing pipeline, splits, fitting protocol, and parameters.
Reproduction results	Reproduction slide: Tissue/MPL reproduction and public-curve baseline validation.
Method and result comparison	Public-curve experiments: MPL vs MPL+$\mathrm{DropRelaxS}$, transfer table, and fair-baseline diagnostic.
Analysis and reflection	Independent reproduction, negative results, limitations, and next-work slides.
Code and work division	Reproducibility map plus explicit code/running/division slide.

Design choice. The slides are not a speaking outline; they are written so that the project can be understood by reading the deck alone.

The problem: predict the whole loss curve before training

Goal. Given a learning-rate schedule $\{\eta_t\}$, predict the pretraining loss curve $L(t)$ without running the full training job.

Why this matters.

- LLM pretraining is expensive; schedule search by full trial runs is wasteful.
- Standard scaling laws mostly target endpoint loss; schedule-aware laws must predict the *trajectory* under warmup, stable phases, cooldowns, and WSD-style drops.
- The hardest test is cross-schedule transfer: fit on slow schedules such as cosine, then predict unseen sharp-decay WSD curves.

Project question.

Do current loss-curve laws only know how much LR was spent, or also how fast it was spent?

Baseline laws: strong, but structurally adiabatic

Let $S(t) = \sum_{i \le t} \eta_i$ be cumulative LR and $\Delta\eta_k = \eta_{k-1} - \eta_k$ be a LR decrement.

Multi-Power Law (MPL).

$$L(t) = L_0 + A\,S(t)^{-\alpha} + B \sum_{k \le t} \Delta\eta_k\, G\big(\eta_k^{-\gamma}(S(t) - S(k))\big)\,, \quad G(x) = 1 - (1 + Cx)^{-\beta}.$$

Functional Scaling Law (FSL). FSL gives a closely related functional predictor over the LR history.

What they capture well.

- Integrated progress through $S(t)$.
- A saturating benefit from past LR drops.
- Cross-schedule structure in public LLM curves.

The gap. Their response to a drop is essentially quasi-static: after fitting the schedule kernel, they do not explicitly model the finite-time tracking error of a moving LR-dependent noise floor. I call this an **adiabatic approximation**.

Core thesis

Loss depends not only on how much LR has been spent, but also on how fast it was spent.

Mechanism.

- At a fixed LR, the optimizer has an equilibrium excess-loss/noise-floor state.
- A sharp LR drop lowers that equilibrium immediately.
- The actual excess-loss state relaxes only over a finite time.
- During this transient, the true loss is **higher** than the adiabatic prediction.

Resulting signature. Slow and fast schedules may reach the same final LR, but the fast sweep retains a positive residual. This is not just a function of cumulative LR S; it is a rate-dependent lag.

Data source and processing pipeline

Public-curve data.

- Source: official MPL public loss curves vendored under `external/MultiPowerLaw/loss_curve_repo/csv_{25,100,400}`.
- Scales: 25M, 100M, 400M.
- Schedules: cosine, constant, WSD, WSD-long-decay, and WSD constant-tail probes.

Processing pipeline.

1. Load per-run CSV curves and learning-rate schedules.
2. Align curves by training step; compute cumulative LR $S(t) = \sum_{i \leq t} \eta_i$ and LR decrements $\Delta\eta_k = (\eta_{k-1} - \eta_k)_+$.
3. Fit baseline laws on selected training curves using log-loss Huber objectives.
4. Evaluate predictions on held-out WSD-family curves using MAE.
5. Compute residuals $L_{\text{true}} - L_{\text{MPL}}$ and fit/test $\mathrm{DropRelaxS}$ only on the residual structure.

Main parameters. MPL has $(L_0, A, \alpha, B, C, \beta, \gamma)$; $\mathrm{DropRelaxS}$ adds λ_{slow}, c, and independently measured $\chi = dL_{\text{eq}}/d\eta$.

Experimental split and evaluation protocol

Course-task reproduction split.

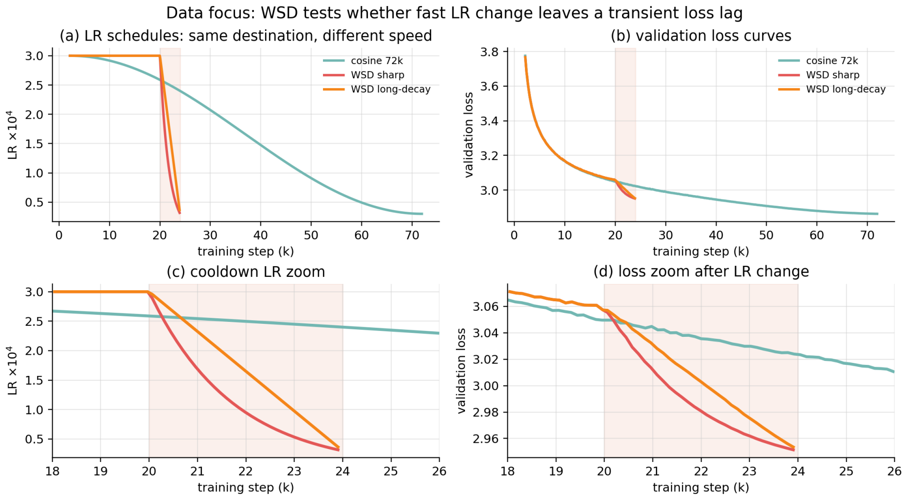
- Training curves: `cosine_24000`, `cosine_72000`.
- Held-out test curves: `wsd_20000_24000`, `wsdld_20000_24000`, `wsdcon_3`, `wsdcon_9`, `wsdcon_18`.
- Evaluation scales: 25M, 100M, and 400M.

Additional public-curve experiments.

- Use a faithfully reproduced MPL as the adiabatic baseline.
- Measure τ from two-stage `wsdcon` constant-tail windows.
- Use `wsd`/`wsdld` as held-out sharp-decay targets for low-calibration cross-scale transfer.

Why this protocol is fair. The correction is evaluated after MPL has already captured the backbone and schedule-drop structure. Therefore any remaining positive residual is not simply a weak baseline artifact.

Data visualization: WSD isolates LR-change speed

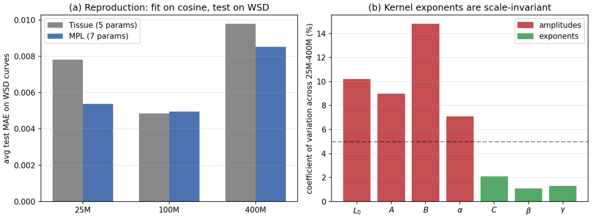


The key contrast is local: WSD reaches low LR much faster, so the cooldown window is where rate dependence should appear.

Reproducing existing methods: Tissue and MPL

Scale	Tissue	MPL
25M	0.00783	**0.00539**
100M	**0.00486**	0.00496
400M	0.00979	**0.00852**

Average WSD-family test MAE.

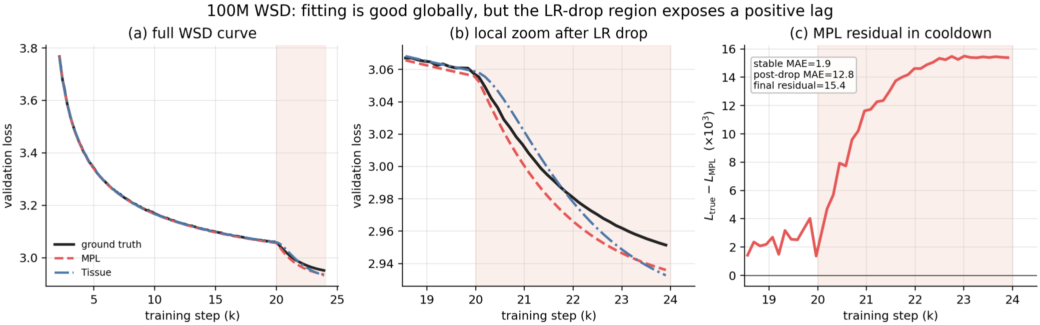


Key differences and difficulties.

- Tissue is compact and interpretable but less stable on sharp WSD families.
- MPL is higher-capacity and matches the published public-curve results.
- Correct optimizer protocol matters: weak or inconsistent fitting can make false “improvements” look real.

Conclusion. The reproduced MPL is the baseline used for all residual analysis.

Curve-level fitting: zoom into the WSD cooldown



Globally the fit is good; locally after the LR drop, MPL predicts too low, leaving a positive residual.

Theory in one slide: linear response of a moving equilibrium

Assume an LR-dependent hidden excess-loss state z_t with local equilibrium $z^\star(S_t, \eta_t)$:

$$z_{t+1} = \Phi_{S_t,\eta_t}(z_t), \qquad L_t = L_{\mathrm{ad}}(S_t, \eta_t) + r_t.$$

Linearizing around the moving equilibrium gives the cooldown-specific residual

$$r_t \approx \sum_{k<t} K_{k,t}\,(\eta_k - \eta_{k+1})_+.$$

Under the first-order optimizer approximation $T_t \simeq I - \eta_t A$, products of transition matrices become exponentials in **cumulative-LR time**:

$$T_{t-1} \cdots T_{k+1} \simeq \exp[-A(S_t - S_k)].$$

Primitive prediction

The non-adiabatic residual is a causal convolution of LR decrements with a stable memory kernel in S-time, not a free bump fitted after the fact.

Operational correction: DropRelaxS

The full kernel can be a spectral mixture. To avoid overfitting, the paper uses the lowest-capacity one-pole closure:

$$\boxed{L(t) = L_{\mathrm{MPL}}(t) + c\,\chi \sum_{k<t} e^{-\lambda_{\mathrm{slow}}(S_t - S_k)}(\eta_k - \eta_{k+1})_+}$$

$$\chi = \frac{dL_{\mathrm{eq}}}{d\eta}.$$

Interpretation.

- $(\eta_k - \eta_{k+1})_+$: only LR drops create the positive lag.
- $e^{-\lambda_{\mathrm{slow}}(S_t - S_k)}$: memory decays in cumulative LR time.
- χ: amplitude equals the equilibrium noise-floor sensitivity.
- c: low-capacity compression of a spectral mixture into one pole.

Important convention. If drops are normalized by η_{peak}, the prefactor must absorb that factor. With raw LR drops, the dimensionally clean amplitude is $c\chi$.

AdamW specialization: expected relaxation scaling

For a noisy quadratic mode with curvature λ_i and gradient-noise standard deviation s_i, noise-dominated AdamW has $v_i^\star \simeq s_i^2$. Locally it behaves like preconditioned SGD:

$$x_{i,t+1} = \left(1 - \eta_t \frac{\lambda_i}{s_i}\right) x_{i,t} - \eta_t \frac{1}{s_i} \xi_{i,t}.$$

For $V_i = \mathbb{E}[x_i^2]$:

$$V_i^\star(\eta) = \frac{\eta s_i}{2\lambda_i} + O(\eta^2), \qquad F_i^\star(\eta) = \frac{1}{2}\lambda_i V_i^\star(\eta) = \frac{\eta s_i}{4} + O(\eta^2).$$

Two useful consistency checks.

- **Amplitude identity:** $\sum_i dF_i^\star/d\eta = \frac{1}{4}\sum_i s_i + O(\eta)$.
- **Classical relaxation time:**

$$\tau_i^{-1} = 2\eta \frac{\lambda_i}{s_i}, \qquad \tau_i \propto \eta^{-1}.$$

This timescale is not a new theorem; here it is used as a mechanistic check for whether the **residual left by adiabatic loss-curve laws** behaves like finite-time relaxation.

Predictions and falsifiability

The correction predicts six observable signatures.

1. Residual after fast LR drops is positive.
2. Residual depends causally on LR decrements, not just on LR level.
3. Memory is in cumulative LR time S, not raw step time.
4. Amplitude is controlled by $dL_{\mathrm{eq}}/d\eta$.
5. In a noise-dominated AdamW regime, residual relaxation should be consistent with $\tau \propto 1/\eta$.
6. A spectral mixture should sometimes beat a single exponential.

Failure modes are also meaningful.

- If a full set of WSD curves is available, MPL can absorb the residual by refitting.
- If the scale is too small/noisy, this relaxation scaling may not be resolved.
- A single decay curve can make finite-λ_{slow} nearly degenerate with the adiabatic limit; rate dependence must be identified across schedules.

Public-curve experimental setting

Data.

- Official MPL public LLM loss curves.
- Llama-2-style models at 25M, 100M, and 400M.
- Nine schedules per scale, including cosine, constant, WSD, WSD-long-decay, and two-stage WSD constant-tail probes.

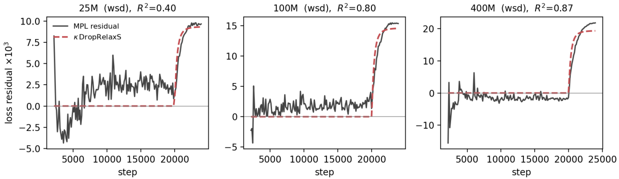
Baseline.

- Use a faithfully reproduced well-fit MPL.
- Test MAE matches Luo et al.: 0.0054/0.0050/0.0085 at 25/100/400M.
- This matters: the residual is measured after a strong adiabatic baseline, not after a weak fit.

Evaluation.

- Residual structure on held-out sharp decays.
- τ measured from two-stage `wsdcon` relaxation transients.
- Cross-scale transfer on held-out `wsd`/`wsdld` curves.

Result 1: fast decays leave a structured positive residual



Observation.

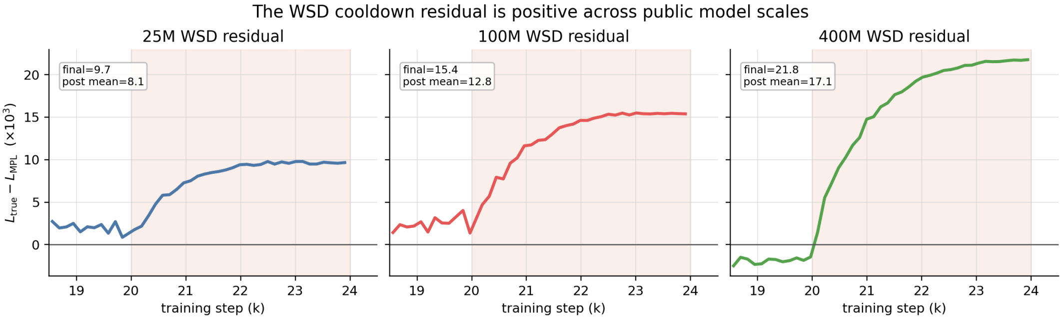
- During the stable phase, the well-fit MPL residual is near zero.
- At a sharp WSD cooldown, the residual jumps positive.
- The one-pole $\mathrm{DropRelaxS}$ term tracks the residual.

Fit quality.

$$R^2 = 0.40/0.80/0.87 \quad (25/100/400\mathrm{M}).$$

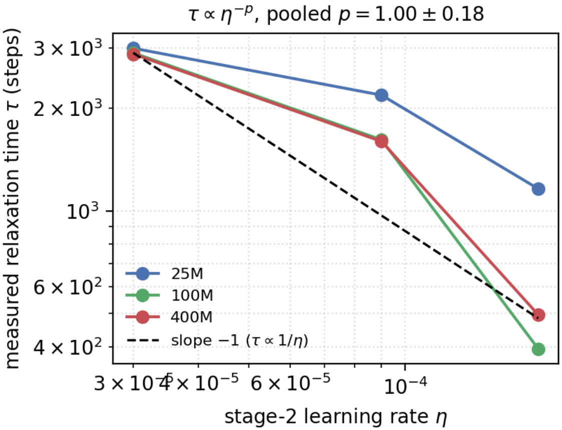
Interpretation. The signal strengthens with scale. The residual is exactly the expected lag: after a fast drop, the true loss stays above the quasi-static prediction.

Result 1b: WSD cooldown residual across scales



The same post-drop sign appears at 25/100/400M, and the residual magnitude is larger at larger scale.

Consistency check: relaxation time vs. LR



Protocol.

- Use two-stage `wsdcon` curves.
- At step 8000, LR drops to a constant second-stage value: $\eta \in \{3, 9, 18\} \times 10^{-5}$.
- Fit the residual transient $r(t) \propto e^{-(t-8000)/\tau}$.

Observed trend.

$$\tau \propto \eta^{-p}, \qquad p = 1.00 \pm 0.18$$

pooled over public curves; 100/400M track the classical slope -1 most cleanly.

Interpretation. This supports the finite-time relaxation mechanism, but it is not the main claim: the main claim is the positive WSD residual and its useful correction.

Result 3: amplitude is tied to the noise floor

Independent amplitude probe. Use final losses of two-stage constant-tail curves to estimate the equilibrium noise-floor sensitivity $dL_{\mathrm{eq}}/d\eta$ without fitting $\mathrm{DropRelaxS}$ to the target curve.

Scale	25M	100M	400M
$\kappa_{\mathrm{fit}}/\kappa_{\mathrm{pred}}$	0.43	0.51	0.60

Analysis.

- The ratio has coefficient of variation 14% across scales.
- The < 1 factor is expected: the single-pole closure compresses a spectral mixture.
- With raw LR drops this gives $c \approx 0.5$; with normalized drops $(\eta_k - \eta_{k+1})_+/\eta_{\mathrm{peak}}$, the same convention requires $c \approx 60$–86.

Practical conclusion

The transferable object is the near-constant ratio under a clearly stated normalization convention, not a universal first-principles value of c.

Result 4: low-calibration cross-scale transfer improves sharp decays

Transfer protocol.

- Fix $\lambda_{\mathrm{slow}} = 10$ from the public-curve relaxation measurement.
- Take c from a leave-one-scale-out average of the other scales.
- Measure $dL_{\mathrm{eq}}/d\eta$ from cheap two-stage probes at the target scale.
- Do **zero fitting on the held-out target curve**.

	`wsd`		`wsdld`		
Scale	MPL	MPL+DropRelaxS	MPL	MPL+DropRelaxS	avg. change
25M	0.00341	0.00246	0.00314	0.00219	−29%
100M	0.00375	0.00164	0.00325	0.00161	−53%
400M	0.00470	0.00236	0.00385	0.00212	−47%
				overall	**−44% (6/6 wins)**

Analysis. The correction is most useful in the data-poor regime: cosine plus cheap noise-floor probes, but no full WSD curves.

Upgrade 1: the B-identity — the amplitude was inside MPL all along

Proposition. Compare holding η vs dropping to $\eta - d\eta$ and equilibrating at matched S: MPL's annealing term changes by $-B\,d\eta\,G(\cdot) \to -B\,d\eta$ as $G \to 1$, so

$$\partial_\eta L_{\mathrm{eq}}^{\mathrm{MPL}}\big|_S = B.$$

Falsifiable check. $B = 363.8/437.9/523.4$ vs independently measured $dL_{\mathrm{eq}}/d\eta = 381.7/499.3/561.5$ (5–15% at all scales).

Zero-extra-measurement transfer

$\kappa = c''\eta_{\mathrm{peak}}B$ (c'' leave-one-scale-out, CV 14.7%) gives $-$**43**% (6/6) on the Table-1 protocol — matching the probe chain at zero additional training runs. Smooth-only MPL refits (B drifts $\leq 30\%$) retain -36 to -40%.

Upgrade 2: second-order amplitude closure $\chi(\eta) \propto \eta^{\delta}$

Measurement. Settled floors are superlinear: $F_{\rm eq} \propto \eta^{1+\delta}$, $1+\delta = 1.06/1.49/1.25$ (25/100/400M). The first-order closure froze χ; the derivation's forcing $\partial_\eta z^\star(S_k, \eta_k)$ already depends on η_k.

$$L(t) = L_{\rm MPL}(t) + \kappa \sum_{k<t} \left(\frac{\eta_k}{\eta_{\rm peak}}\right)^{\delta} e^{-\lambda_{\rm slow}(S_t - S_k)} \frac{(\eta_{k-1}-\eta_k)_+}{\eta_{\rm peak}}$$

- Public curves, $\delta{=}1/4$ (Pareto-selected, never target-fit): audited $6{\times}6$ matrix improves on *every* aggregate; probes-only calibration $-17\% \to -23$ to -29%.
- Real 10.7M model, never-scanned out-of-family beds: `sharp600` $-18\% \to -71\%$ ($\delta{=}3/4$); showcase `wsd_sharp` $-14\% \to -52\%$.
- Honest split: part of the cheap-probe gain is amplitude recalibration; at matched amplitude the unweighted shape stays best in-family. Naive NQM route falsified at magnitude; bulk+edge spectrum (tracking cutoff $a_{\max} \propto 1/\eta$) is the consistent mechanism (conjecture flagged).

Fair-baseline diagnostic: formula gap vs information gap

Key question. Is $\mathrm{DropRelaxS}$ a better formula, or does it just supply information missing from a few-shot MPL fit?

Diagnostic.

- If MPL is refit on several full WSD curves, it can absorb the non-adiabatic residual.
- In that data-rich setting, the explicit $\mathrm{DropRelaxS}$ term adds little.
- In the data-poor setting, $\mathrm{DropRelaxS}$ provides a structured prior that transfers from cheap probes and other scales.

Honest scope

The correction is an **information-efficient leading-order term**. It is not meant to beat a fully refit, data-rich MPL on its own training distribution.

Calibration audit: staying within the current law

Question. Can the current law be made more reliable by changing only how λ_{slow} and the amplitude are calibrated?

Fixed model class.

$$L(t) = L_{\mathrm{MPL}}(t) + \kappa\,\mathrm{DropRelaxS}_\lambda(t), \qquad \mathrm{DropRelaxS}_\lambda(t) = \sum_{k<t} e^{-\lambda(S_t - S_k)}(\eta_k - \eta_{k+1})_+.$$

Calibration protocol	Best λ	Final WSD MAE change	wins
fit κ on `wsdcon_9` probe	10	-14.5%	6/6
fit κ on all `wsdcon` probes	7–10	-17.9%	6/6
cross-scale c + noise-floor sensitivity	5–10	-44.0% to -45.0%	6/6

Interpretation.

- The current law improves final cosine-fit MPL $\to$ WSD prediction under several non-target calibration choices.
- $\lambda = 10$ is not a hand-tuned optimum; it is close to the best range and is the independently measured value from the relaxation probe.
- The large 44% gain relies on the stronger cross-scale amplitude chain, not merely on fitting residuals on one probe curve.

Final kappa estimator: remove nuisance drift before transfer

Problem with raw amplitude fitting. The residual after MPL contains both the schedule-response feature and smooth MPL drift. Directly fitting κ can get the trend direction right but the magnitude wrong.

Final calibration model.

$$r = L_{\mathrm{obs}} - L_{\mathrm{MPL}} = \kappa_\star \phi + g_\star + \varepsilon, \qquad g_\star \in \mathcal{G}, \quad \kappa_\star \geq 0.$$

Let $M_{\mathcal{G}}$ residualize against the low-frequency nuisance subspace:

$$\phi_\perp = M_{\mathcal{G}} \phi, \qquad r_\perp = M_{\mathcal{G}} r, \qquad R = \frac{\|\phi_\perp\|_2^2}{\|\phi\|_2^2}.$$

Cap-free paper-facing estimator.

$$\widehat{\kappa} = \sqrt{R} \left(\frac{\langle \phi_\perp, r_\perp \rangle}{\|\phi_\perp\|_2^2 + \tau^2} \right)_+, \qquad \tau = \sigma / k_0.$$

Interpretation: FWL partial regression + empirical-Bayes ridge + identifiable-amplitude conversion.

Final kappa audit: what changed

Cross-schedule transfer matrix.

Estimator	worst offdiag	mean offdiag	cosine→WSD	max cosine κ
smooth baseline	-0.0%	-10.1%	-0.0%	0.0000
EB without nuisance control	-1.0%	-10.8%	-3.1%	0.0045
final cap-free	-2.7%	-12.1%	-4.3%	0.0089
spectral G_4 audit	-1.8%	-10.0%	-3.6%	0.0070
no identifiable conversion	$+1.8\%$	-12.2%	-17.4%	0.0300

Readout.

- The DCT low-frequency audit is slightly more conservative, but shows the result is not tied to a polynomial-shaped nuisance basis.
- The hard cap is not the mechanism: cap-free and capped legacy variants have the same worst off-diagonal result.
- Removing $\sqrt{R}$ creates a positive off-diagonal failure.
- Train-only τ preserves the conclusion: worst remains -2.7%, mean remains -12.1%, and cosine→WSD becomes -5.6%.

Scripts: `repro/current_law_final_kappa.py`, `repro/current_law_trainonly_tau_audit.py`.

How to choose the nuisance subspace

The object is low-frequency drift control, not a polynomial fit. Use a DCT low-frequency family:

$$\mathcal{G}_K = \mathrm{span}\{q_0, \ldots, q_K\}, \qquad R_K = \frac{\|M_{\mathcal{G}_K}\phi\|_2^2}{\|\phi\|_2^2}.$$

Automatic spectral rule.

$$K^\star = \arg\min_{K \geq K_{\min}} |R_K - \rho|.$$

Choice of $\mathcal{G}$	worst offdiag	mean offdiag	cosine$\to$WSD
fixed DCT $K = 4$	-1.8%	-10.0%	-3.6%
retention target only	$+304.0\%$	$+8.4\%$	-9.5%
$K_{\min} = 3, \rho = 0.35$	-1.7%	-11.2%	-10.1%

Conclusion. First enforce minimum low-frequency drift control; then use identifiable feature energy to choose among sufficiently rich nuisance spaces.

Split audit: where the current law fails

Research question. If we calibrate λ and κ on arbitrary non-cosine curves, does the same law generalize to the rest?

Audit	Result	wins	Interpretation
all non-cosine train/test splits	median -14.1% MAE	182/225	usually helpful
probe-only $\rightarrow$ final `wsd`/`wsdld`	mean -14.8% MAE	42/42	robust target use case
sharp decays $\rightarrow$ `wsdcon` tails	up to $+96\%$ MAE	fail	over-corrects long tails

Conclusion. $\mathrm{DropRelaxS}$ should not be sold as a universal residual smoother. Its defensible role is narrower: a leading-order sharp-cooldown correction whose amplitude is best fixed by probe curves and cross-scale transfer, then judged on cosine-fit MPL $\rightarrow$ WSD prediction.

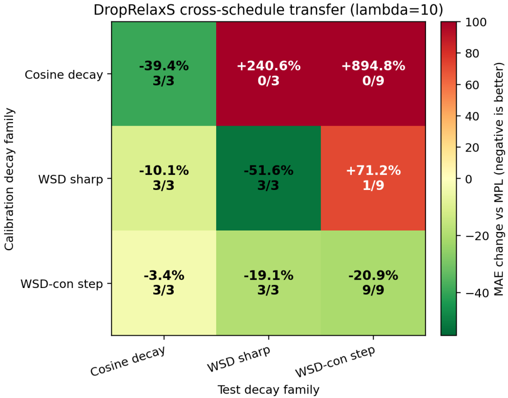
More calibration data: matched beats mixed

Question. If we allow more calibration curves, does the same law become more reliable?

Calibration data	λ choice	Held-out sharp MAE change	wins
other sharp curve only	fixed 10	−49.0%	6/6
other sharp curve only	selected on calibration	−49.9%	6/6
all `wsdcon` probes + other sharp	fixed 10	−19.5%	6/6
all `wsdcon` probes only	fixed 10	−17.6%	6/6

Research conclusion. More data is not automatically better. A full sharp-decay calibration curve transfers strongly to the other sharp shape, but mixing it with long `wsdcon` tails dilutes the amplitude. For the strict no-target-WSD setting, the cross-scale probe rule remains the cleaner claim.

Cross-schedule fitting matrix



Cell text: mean MAE change, wins/tests across 25M/100M/400M and selected curves.

Reading. Cosine calibration does not transfer to WSD targets; `wsdcon` step probes transfer to sharp WSD with smaller but stable gain; sharp-WSD calibration over-corrects `wsdcon` tails. The law is schedule-family dependent, not universal.

Method comparison: what each model can and cannot express

Tissue. Simple LR-integral baseline with few parameters. It is useful as a lightweight reference, but has a coarse schedule response and weaker WSD transfer.

MPL. Strong adiabatic schedule-aware predictor. Public results are reproducible, but lag is absorbed only when full WSD curves are available.

MPL+DropRelaxS. Adds a causal finite-time lag term with low calibration cost. It is most useful for data-poor sharp-decay prediction; constants are still measured and the single-curve degeneracy remains.

Empirical comparison.

- On the course split, MPL reproduces or improves the existing baselines at 25/400M and ties Tissue at 100M.
- On held-out sharp decays, low-calibration MPL+DropRelaxS reduces MAE by 44% over well-fit MPL in the public same-architecture setting.
- When full WSD curves are available, refit MPL absorbs much of the residual; the gain of DropRelaxS is specifically in the "predict before training" regime.
- Exhaustive split audits show the current law can over-correct `wsdcon` tails when calibrated only on sharp decays, so its scope is sharp-cooldown prediction, not arbitrary residual smoothing.

Why an independent reproduction was needed

Problem. The original project data are not fully recoverable: the author-side public loss-curve repository is no longer hosted.

Reproduction strategy.

1. Build a controlled noisy-quadratic AdamW simulator from scratch.
2. Train an independent ∼10M byte-level transformer on WikiText.
3. Run adversarial audits: check for estimator failures, degeneracies, leakage, and windowing artifacts.

Purpose. The reproduction is not just a rerun. It tests whether the mechanism survives outside the original public curves and clarifies which quantitative claims do *not* transfer at small scale.

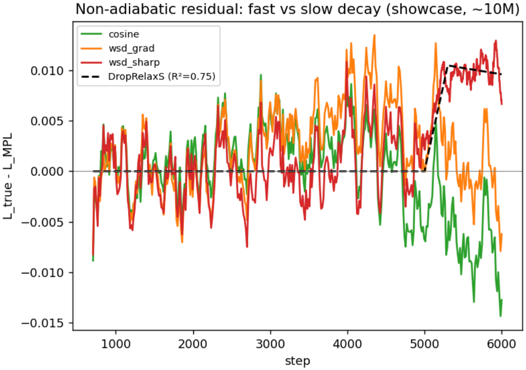
Real ∼10M transformer: no-fitting evidence of rate dependence

Schedule	final loss	cumulative LR S
cosine (gradual)	1.0490	8.20
wsd (gradual)	1.0482	9.22
wsd (sharp)	**1.0593**	**12.59**

All schedules reach the same final LR.

Analysis.

- Sharp sweep: higher loss despite 37% more cumulative LR.
- Any law monotone in S predicts the opposite.
- Across schedules, finite-λ_{slow} $\mathrm{DropRelaxS}$ beats the adiabatic limit: $\Delta R^2 \approx +0.18$.



Controlled AdamW simulation: quantitative mechanism checks

Simulator. Noisy quadratic model with real AdamW updates: true m and v EMAs, true $m/(\sqrt{v}+\epsilon)$ preconditioning, Gaussian gradient noise, and Monte-Carlo ground truth. Closed-form linear theory is used only as an independent baseline.

Claim tested	Result	Interpretation
τ vs LR	$p = 0.971$, log-log $R^2 = 0.9996$	consistent noise-dominated scaling
τ vs β_2	CV $= 1.1\%$ over $[0.9, 0.9999]$	lag not an EMA artifact
Amplitude identity	3.075 vs closed-form 3.0	within 3%
Spectral mixture	recover rates $\{0.87, 16.1\}$ for true $\{1, 16\}$	two-pole structure real

Falsifiability. A signal-dominated spectrum breaks the slope-1 pattern, so the check is not a circular fit to a fixed exponent.

What does not transfer at $\sim$10M

Negative results are part of the final claim.

- The real $\sim$10M transformer does **not** resolve $\tau \propto 1/\eta$: over a $16\times$ LR range, measured post-step relaxation is roughly flat ($\tau \approx$ 475–912 steps, $p \approx 0.18$).
- Audits reject the simple "window truncation" excuse: synthetic recovery tests and profile-likelihood checks reject very large hidden τ values.
- On a single decay curve, finite-λ_{slow} $\mathrm{DropRelaxS}$ is numerically close to the adiabatic limit (correlation ≈ 0.99). Rate dependence is identifiable only across schedules.
- Frozen cross-shape transfer in controlled simulations fails its pre-registered threshold (mean $R^2 = 0.52 < 0.6$).

Bounded conclusion

The qualitative non-adiabatic lag is robust. The inverse-LR relaxation trend is useful supporting evidence in public larger curves and controlled AdamW, but it is not established by the small real transformer.

What is solid

Main evidence.

- Fast LR decays create a positive residual relative to well-fit adiabatic laws.
- The residual has the predicted causal LR-decrement structure.
- Public LLM curves show residual relaxation consistent with inverse-LR scaling ($p = 1.00 \pm 0.18$), as a mechanism check rather than the main claim.
- Controlled AdamW reproduces the same trend, β_2 robustness, amplitude identity, and spectral-mixture behavior.
- A real $\sim$10M transformer shows no-fitting rate dependence: the fast sweep ends higher despite more cumulative LR.

Practical value. In data-poor sharp-decay prediction, low-calibration $\mathrm{DropRelaxS}$ transfer reduces held-out MAE by 44% on public curves (6/6 wins).

What remains open

Constants are measured, not solved.

- λ_{slow} is best interpreted as a slow preconditioned mode $\sim 2(\lambda/s)_{\mathrm{eff}}$, not a universal spectral average.
- A robust first-principles predictor still needs an unexplained $O(1)$ constant.
- The normalization of LR drops must be stated explicitly; otherwise $c \approx 0.5$ and $c \approx 60$–86 can appear contradictory while describing the same data under different conventions.

Scope limits.

- Real-data public curves cover one architecture family and three scales.
- The $\sim$10M independently trained model confirms the effect but not a clean inverse-LR relaxation trend.
- The relationship to MPL's empirical exponent γ is complementary and still only qualitative.

Next work: highest-value experiments

Priority 0: strengthen current-law calibration.

- Add more two-stage probe LRs to estimate χ and λ_{slow} with tighter confidence intervals.
- Pre-register $\lambda = 10$ and the cross-scale c rule, then evaluate on new WSD shapes.
- Test whether the same c transfers across architectures or only within one family.

Priority 1: real-model amplitude identity.

- Train 3–4 constant-LR curves at $\sim$10M.
- Independently estimate $dL_{\text{eq}}/d\eta$.
- Compare it against the DropRelaxS fitted amplitude.

Priority 2: S-time control.

- Alter batch size or effective LR integration so step-time and S-time are separated.
- Directly test whether memory follows $S_t - S_k$ rather than $t - k$.

Priority 3: scale and spectrum.

- If compute allows, repeat the real-transformer relaxation probe at $\geq$ 25M.
- If diagnostics are available, measure Hessian/noise spectra and test the slow-mode explanation for λ_{slow}.

Reproducibility map

Result	Script / document
MPL reproduction and public-curve baseline	`repro/reproduce_cosine_to_wsd.py`
Residual structure and $\mathrm{DropRelaxS}$ regression	`repro/nonadiabatic_theory.py`
Relaxation-time check on public curves	`repro/deep_tau.py`
S-time vs step-time kernels	`repro/deep_stime.py`, `deep_stime2.py`
Spectral mixture diagnostics	`repro/deep_rigor.py`
Low-calibration cross-scale transfer	`repro/deep_predict.py`
Fair-baseline check	`repro/correction_fair.py`
Current-law calibration audit	`repro/current_law_calibration_search.py`
Current-law split research	`repro/current_law_split_research.py`
More-data calibration research	`repro/current_law_more_data_calibration.py`
Independent NQM simulations	`represent/repro/E*.py`, `represent/repro/G*.py`
Real $\sim$10M transformer training	`represent/repro/train*.py`
Independent audit and final report	`represent/results/AUDIT_PARTC.md`, `represent/REPORT.md`

Dependencies: `pip install -r requirements.txt`.
Main paper: `paper/main.tex`. Independent reproduction: `represent/REPORT.md`.

Code, running instructions, and work division

Repository.

`https://github.com/wjjpku/DL-final`

Minimal running instructions.

1. Install dependencies: `pip install -r requirements.txt`.
2. Reproduce main public-curve results: `python3 repro/reproduce_cosine_to_wsd.py -scales 25 100 400`.
3. Run key non-adiabatic diagnostics: `python3 repro/deep_tau.py`, `python3 repro/deep_predict.py`, `python3 repro/nonadiabatic_theory.py`.
4. Build paper and slides: `pdflatex paper/main.tex`; `pdflatex slides/main.tex`.

Work division. This is a single-member project. Jiaju Wu completed data processing, baseline reproduction, theory derivation, new correction design, experiments, independent reproduction analysis, writing, and slide preparation. Third-party code/data dependencies are kept under `external/` and cited in the paper.

Final takeaway

Learning-rate schedules are not adiabatic.

Substantive conclusion. MPL/FSL-style laws capture strong quasi-static structure, but fast LR decay leaves a finite-time relaxation lag. The leading correction is a causal convolution of LR drops with a memory kernel in cumulative LR time.

Practical conclusion. $\mathrm{DropRelaxS}$ is useful as a low-calibration prior when full WSD curves are unavailable, especially for sharp-decay extrapolation.

Scientific boundary. The mechanism is real; the exact constants are not yet first-principles solved. The honest open problem is to compute λ_{slow} and *c* from spectra, and to test the relaxation trend on larger independently trained models.